

Linear Contexts, Tables, Equations, and Graphs

Grade 8 / Pre-Algebra

Read a table or graph, find the rate of change, and build the equation that matches the story.

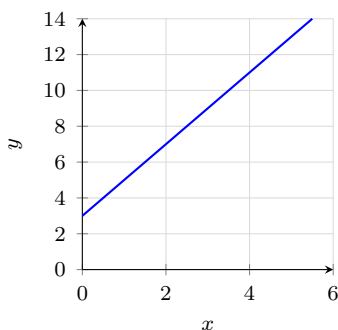
The same linear relationship can arrive as a story, a table, an equation, or a graph. Every problem below asks you to move between two of those forms, so decide first *which* form you have and *which* you need.

- **Rate of change** (slope) = $\frac{\text{change in } y}{\text{change in } x}$ — from any two points of the relationship, however they are given.
- **Initial value** (the y -intercept) is the value when x is zero: where the graph crosses the vertical axis, or the row of the table with a zero input.
- **Equation:** $y = (\text{rate}) \cdot x + (\text{initial value})$. Use it forwards to predict a y , and solve it backwards to find an x .

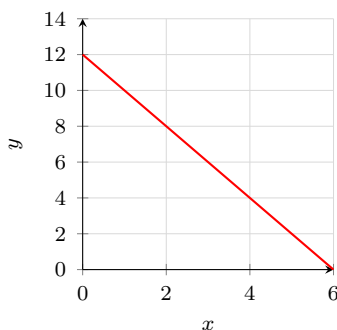
Show the two points or the substitution you used, then write the answer on the answer line.

Reference graphs. Three lines are drawn once here and are called by letter in the problems below. Read the points you need off the grid.

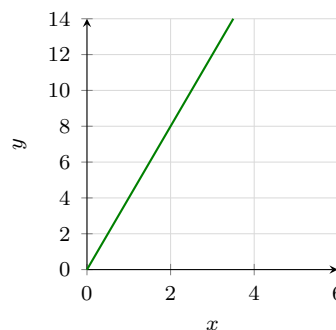
Graph A



Graph B



Graph C



1. A taxi company posts this fare table. The distance is in miles and the fare is in dollars.

Miles	1	2	3	4
Fare	9	14	19	24

- (a) What is the fare for a ride of 3 miles?

Answer: _____

(b) What is the rate of change of the fare, in dollars per mile?

Answer: _____

2. Find the rate of change (the slope) of the line shown in **Graph A**. Write down the two points you read off the grid before you compute.

Answer: _____

3. A gym charges a joining fee of 20 dollars, then 15 dollars for every month of membership. Let m be the number of months.

Write an equation for the total cost, then use it to find the total cost after 8 months. Give the cost in dollars.

Answer: _____

4. Find the rate of change (the slope) of the line shown in **Graph B**. Write down the two points you read off the grid before you compute.

Answer: _____

- 5.** The line in **Graph C** has the equation $y = 4x$.
For what value of x does this line reach $y = 28$?

Answer: _____

- 6.** A tank is draining. The table shows the volume of water left, in litres, after each minute.

Minutes	0	1	2	3
Litres	60	48	36	24

- (a) How much water is left after 2 minutes?

Answer: _____

- (b) What is the rate of change, in litres per minute?

Answer: _____

- 7.** One of the three reference graphs is the graph of the equation $y = 12 - 2x$.
Substitute $x = 5$ into that equation and write the resulting y -value on the answer line. Then name the graph the equation matches, and explain how the point you found proves it.

Answer: _____

8. A different tank is filling at a steady rate. After 3 minutes it holds 26 litres; after 8 minutes it holds 51 litres.

Find the rate at which the tank is filling, in litres per minute.

Answer: _____

9. A seedling is 12 centimetres tall and grows 3 centimetres each week. Let w be the number of weeks from now.

Write an equation for the seedling's height, then use it to predict the height after 9 weeks.

Answer: _____ cm

10. The table below shows a linear relationship.

x	0	2	4	6
y	7	13	19	25

(a) Read the value of y when x is 4.

Answer: _____

(b) Find the rate of change of this relationship.

Answer: _____

11. Compare two representations. **Graph A** and **Graph C** are drawn on identical grids.

Find the rate of change of **Graph C**, then state which of the two lines climbs faster and how you can see that on the grids without computing anything.

Answer: _____

12. Challenge. A phone plan charges a fixed monthly fee plus a charge for each gigabyte of data used. Two months of bills are shown, with data in gigabytes and the bill in dollars.

Gigabytes	2	5
Bill	31	46

(a) Find the charge per gigabyte.

Answer: _____

(b) Build the equation for the bill, then find how many gigabytes give a bill of 61 dollars.

Answer: _____